

Unit 11, Lesson 2: Constructing Line Graphs

Benchmark

MA.8.DP.1.1 Given a set of real-world bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

Learning Target

☐ I can construct a line graph given a set of real-world data.

11.2.1 Warm-Up: Trends

1. The term **trend** is used in many contexts.
 - a. Describe a current fashion trend.
 - b. Describe what it means for something to be “trending” on Netflix or Twitter.
 - c. What do you think “trend” means in mathematics in the context of numerical data?

11.2.2 Guided Instruction: Constructing Line Graphs

1. As of the 2020-2021 NBA season, Jimmy Butler of the Miami Heat played 10 seasons in the NBA. The table shows his average assists per game for his first 10 seasons. In basketball, a player records an assist when he or she passes the ball to a teammate and that teammate scores a field goal (two-point or three-point shot).

Season (Years)	1 (2011-12)	2 (2012-13)	3 (2013-14)	4 (2014-15)	5 (2015-16)
Assists per Game	0.3	1.4	2.6	3.3	4.8
Season (Years)	6 (2016-17)	7 (2017-18)	8 (2018-19)	9 (2019-20)	10 (2020-21)
Assists per Game	5.5	4.9	4.1	6.0	7.3

- a. Identify the independent and dependent variable in this context.

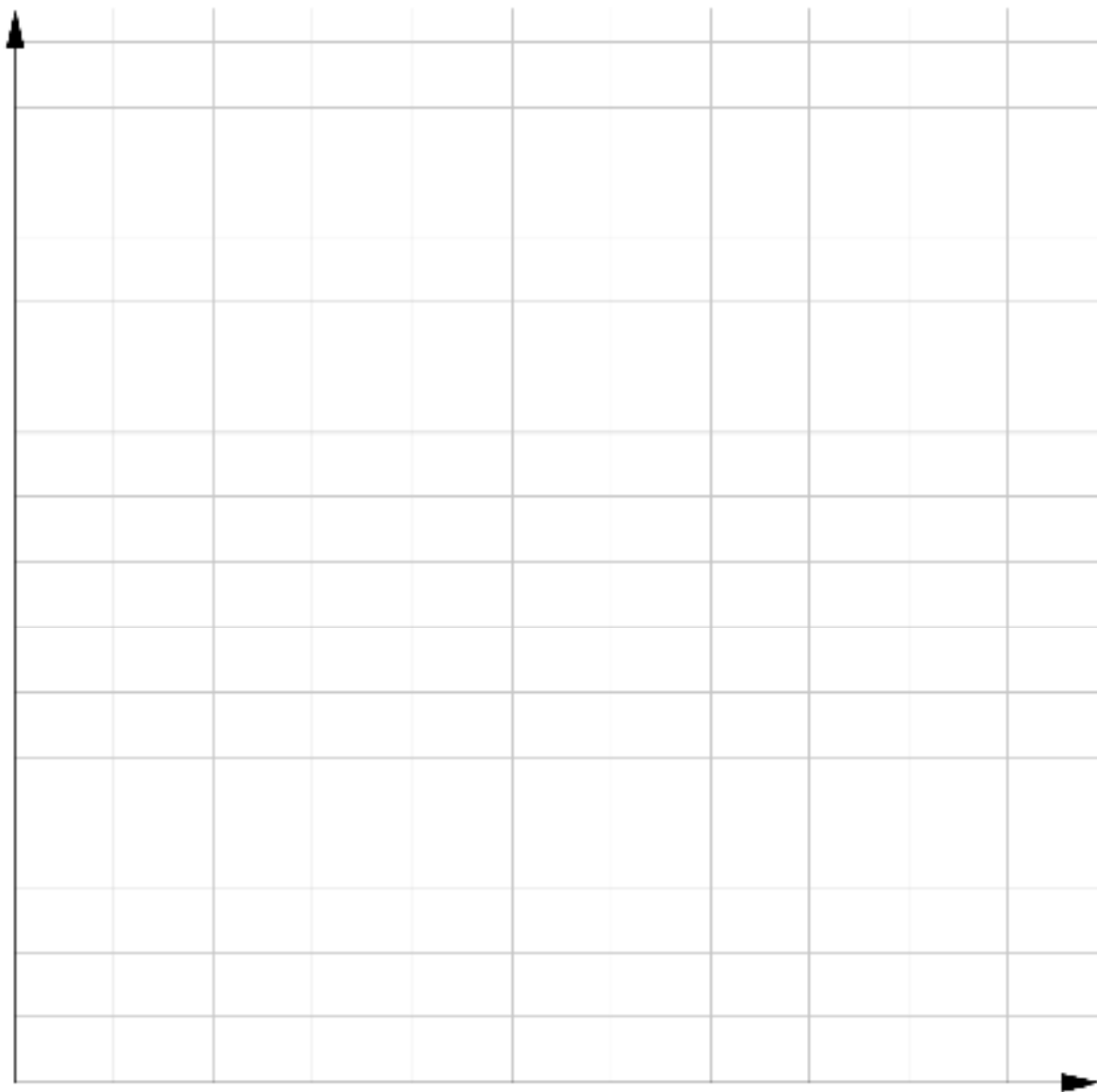
Independent Variable	
Dependent Variable	

On a line graph, the independent variable is represented along the x -axis and the dependent variable is represented along the y -axis.

- b. What range of data values need to be represented along each axis to represent this data set?

Range of x -values	
Range of y -values	

- c. **Construct a line graph to represent the data by completing the following.**
- **Label the axes with the appropriate variables.**
 - **Identify and label an appropriate scale based on the data for each axis on the grid.**
 - **Plot a point for each pair of values in the data set.**
 - **Connect the points in order, using a straightedge.**
 - **Title the graph.**

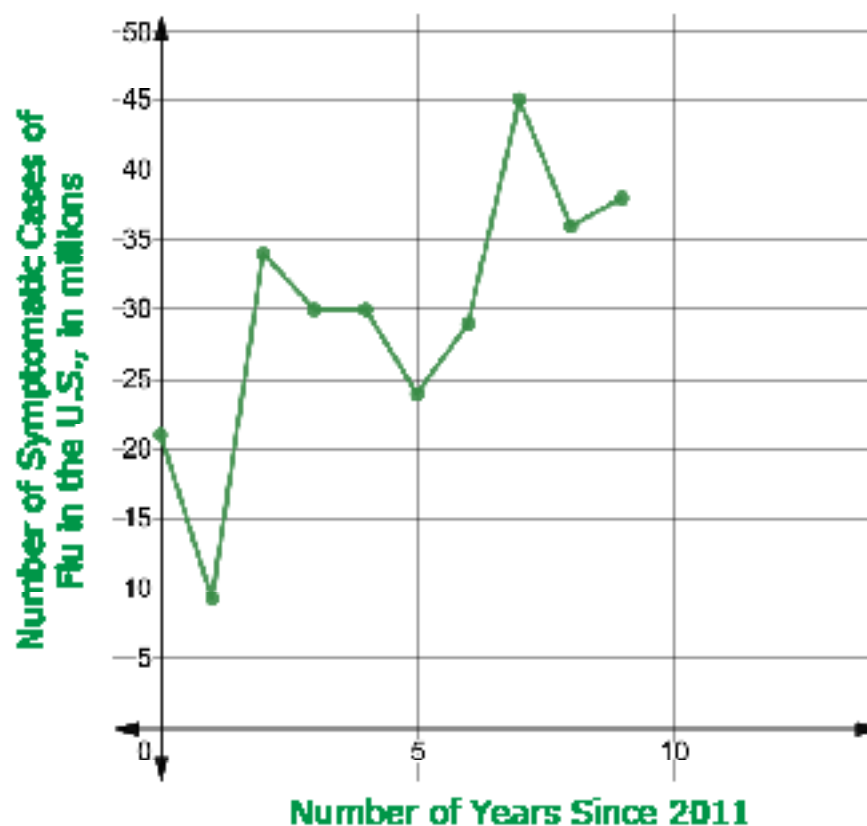




A **trend** in mathematics describes a change over time. In cases of bivariate data where one of the variables is not time, trends describe how one variable changes in relation to changes in the other.

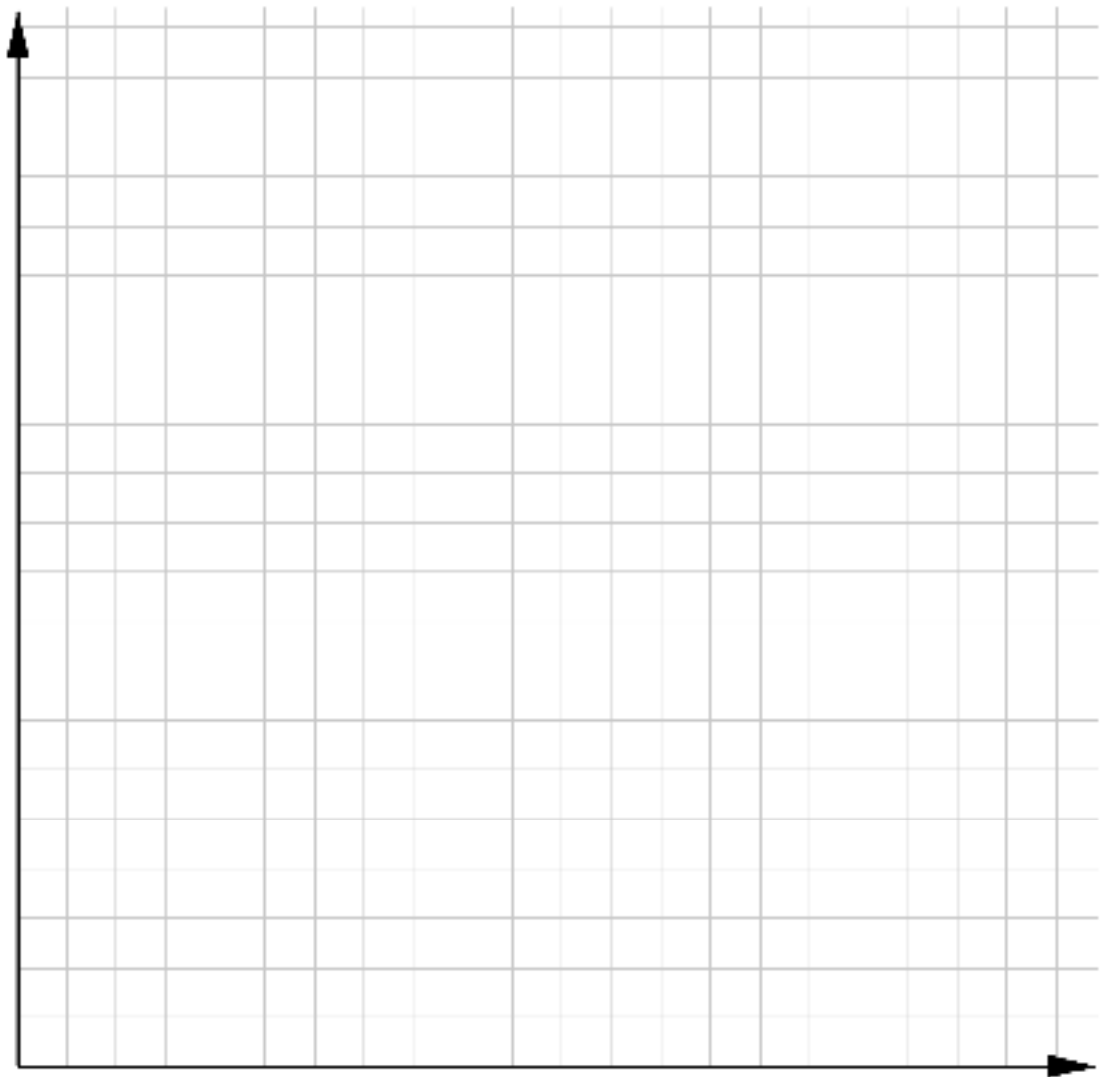
d. Describe the trend of the data represented on the line graph.

2. A line graph from Lesson 1 is shown. The original data set was given in years such as 2011, 2012, 2013, etc. Discuss with your partner how these values were represented on the x -axis of the graph.



11.2.3 Activity: Gallery Walk

1. **Work with your partner or small group to construct a line graph to represent the data set assigned. Be sure to include the following when constructing your graph.**
 - **Label the axes with the appropriate variables.**
 - **Identify and label an appropriate scale based on the data for each axis on the grid.**
 - **Plot a point for each pair of values in the data set.**
 - **Connect the points in order, using a straightedge.**
 - **Title the graph.**



11.2.4 Wrap-Up: Ticket Out the Door

1. Use the data shown to construct a line graph representing the average monthly chance of precipitation, represented as a percentage (%), in Jacksonville, FL.

Month	1	2	3	4	5	6
Chance of Rain (%)	26%	28%	27%	24%	29%	47%
Month	7	8	9	10	11	12
Chance of Rain (%)	57%	52%	43%	25%	18%	21%

